

CIS 580

Machine Perception

Or Geometric Computer Vision

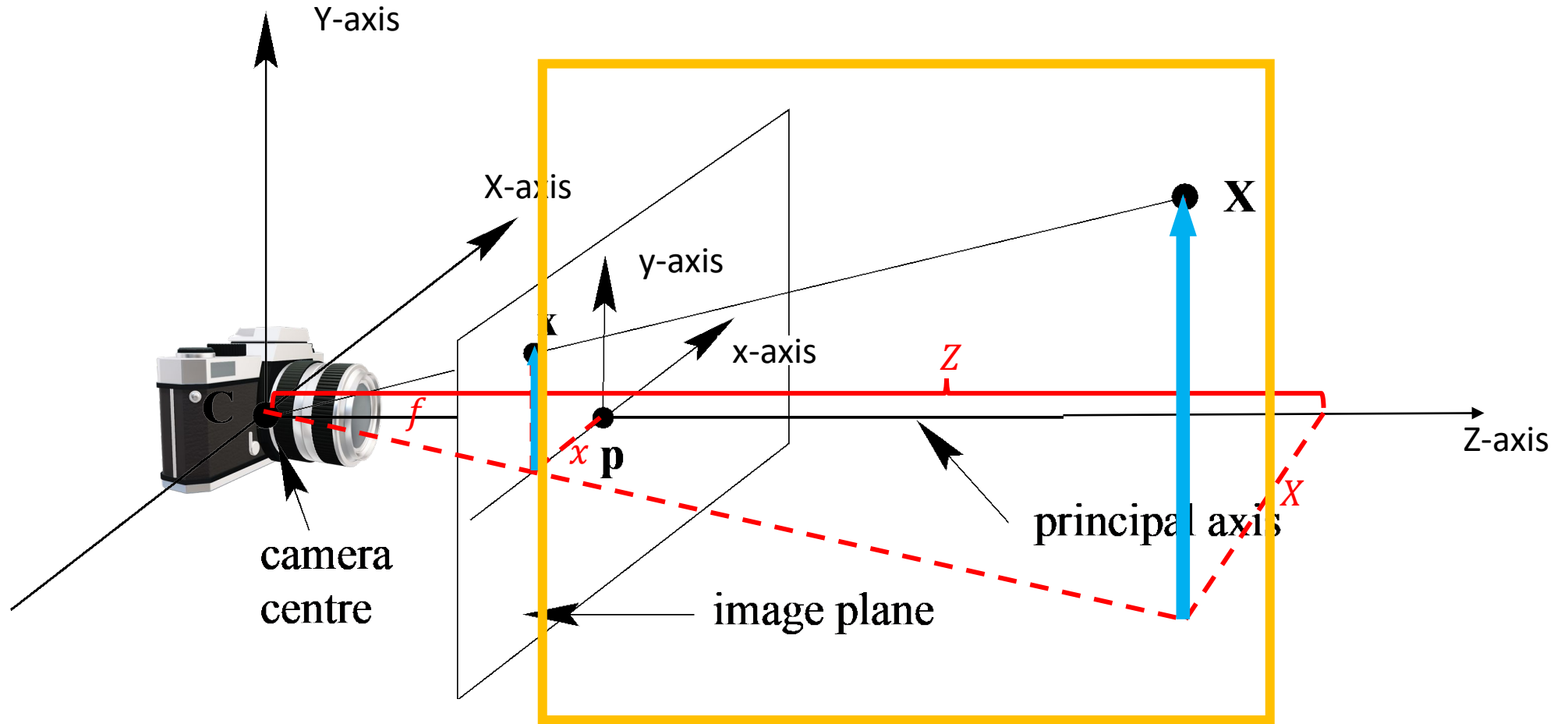
Instructor: Lingjie Liu

Lec 3: Jan 29, 2026

We will start OH this week on Zoom

	Time	Space
Dhyey Shah	Monday 3pm-5pm	Levine Hall Room 612
Siyu Li	Sunday 12-2pm	Levine Hall, Moore 103
Prakriti Prasad	Monday 12-1 pm	Levine GRW 5th floor bump space
Abhimanyu Suthar	Friday 4pm - 6 pm	Levine GRW 3rd floor bump space
Yiming Huang	Wednesday 10 AM - 12 PM	https://upenn.zoom.us/j/8013153196 (Levine GRW 5th floor bump space once returned to campus))
Ravi Raghavan	Thursday 4 - 6 PM	Levine Hall Rm 512
Nandagopal Vidhu	Thursday 11 AM - 12 PM (additional OH on Thursday 10 AM - 11 AM during HW submission week)	Levine Rm 612
Jie Wang	Tuesday 11-12 AM	Levine Hall Room 459
Haoran Yang	Monday 1-3 pm	Levine 601 Bump space
Qihao Yun	Monday 7pm - 8pm	Levine 5th floor Bump Space
Aanal Maheshbhai Sonara	Monday 5pm - 6pm	Levine 501 Bump Space
Yongzhe Wang	Friday 12:30 to 1:30	Levine; CIS Exclusive; Levine 5 Bump Space
Lingjie Liu	Friday 11am - 12pm	Levine 570

Recap: Basic Perspective Projection Equations

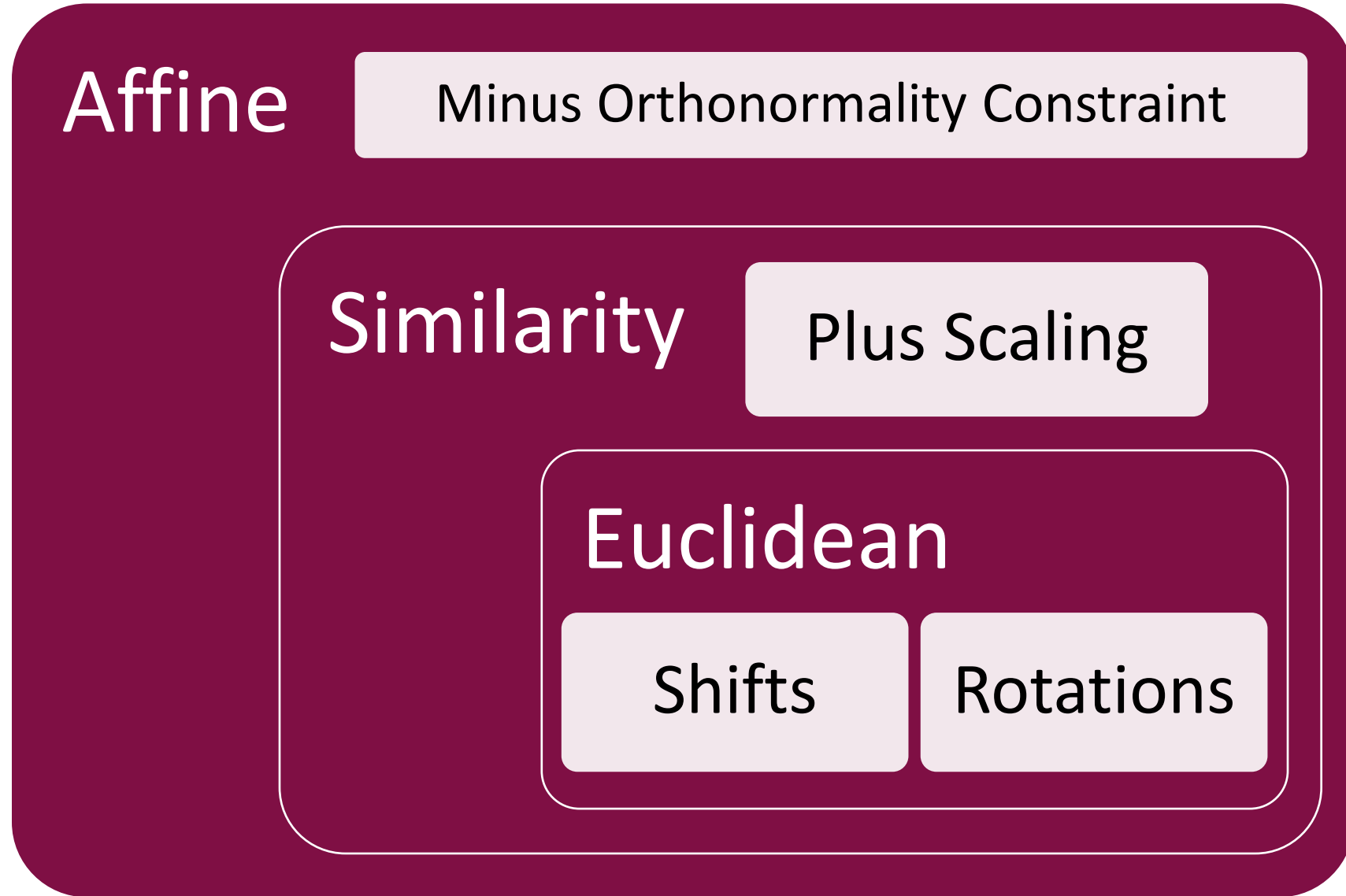


$$x = f \frac{X}{Z}, y = f \frac{Y}{Z}$$

Recap: Points and Planes in Euclidean 3D (shortcut $\mathbb{R}^3$)

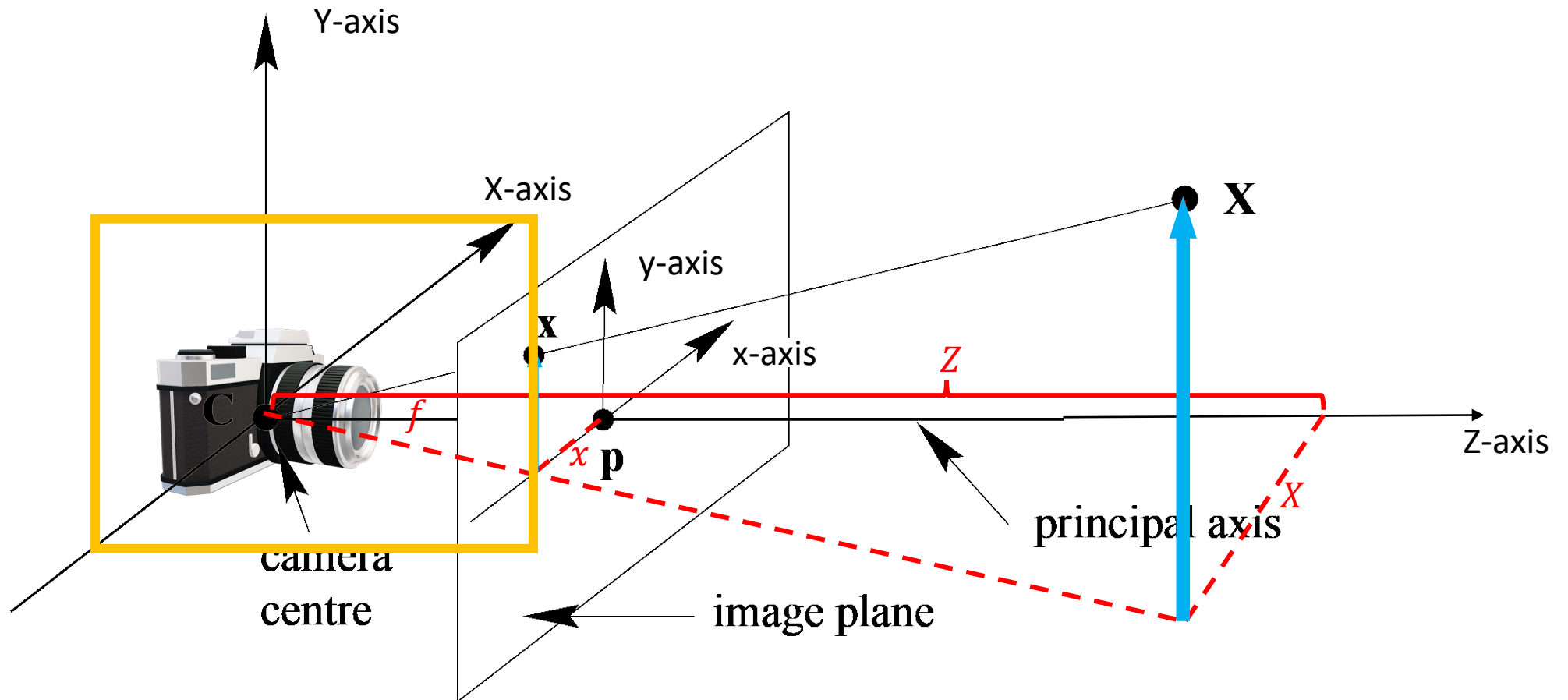
- Point in 3D space: $\mathbf{x} = (x, y, z)^T \in \mathbb{R}^3$
- Plane in 3D space:
 - $ax + by + cz + d = 0$
 - $\boldsymbol{\pi}^T \begin{pmatrix} x \\ y \\ z \\ \mathbf{1} \end{pmatrix} = 0$, where $\boldsymbol{\pi} = [a, b, c, d]^T$
- Line in 3D space:
 - $(x, y, z)^T = (x_1, y_1, z_1)^T + \lambda(x_2 - x_1, y_2 - y_1, z_2 - z_1)^T$ for some value of λ
 - $\begin{pmatrix} \boldsymbol{\pi}_1^T \\ \boldsymbol{\pi}_2^T \end{pmatrix} \begin{pmatrix} x \\ y \\ z \\ \mathbf{1} \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$

Recap: Hierarchy of Transformations



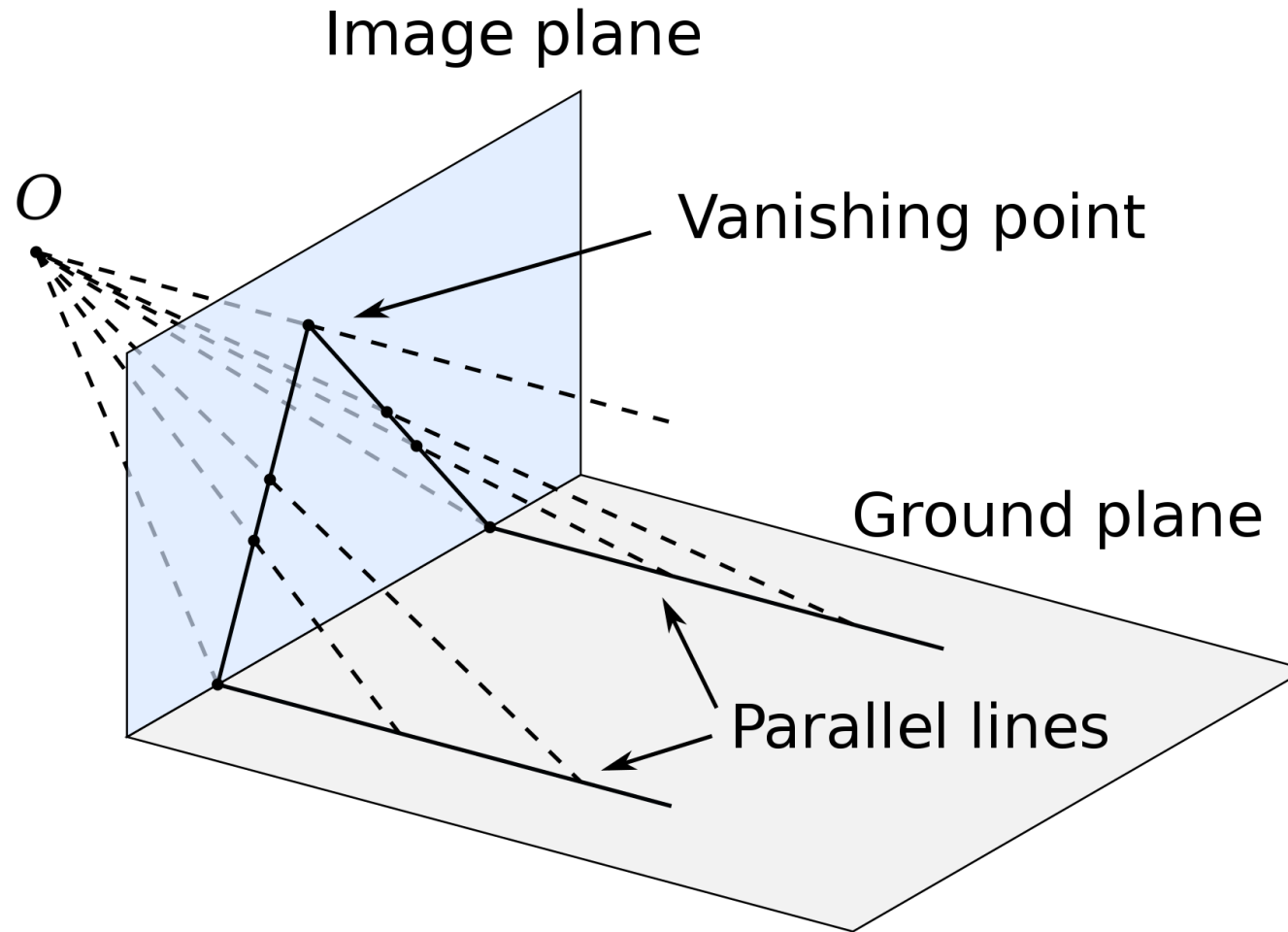
Soon an even more general transformation (“projective”)!

Recap: Basic Perspective Projection Equations



$$x = f \frac{X}{Z}, y = f \frac{Y}{Z}$$

Recap: Where vanishing points come from



Recap: Projective Geometry

- Extension of Euclidean geometry that deals with points at infinity
 - Other than that, Projective Geometry keeps many of the same features as Euclidean geometry.
- A key property is that two lines always meet in a point
 - (sometimes a point at infinity)

Recap: “Homogeneous coordinates”: Euclidean $\mathbb{R}^n \rightarrow$ Projective $\mathbb{P}^n$

Homogeneous coordinates

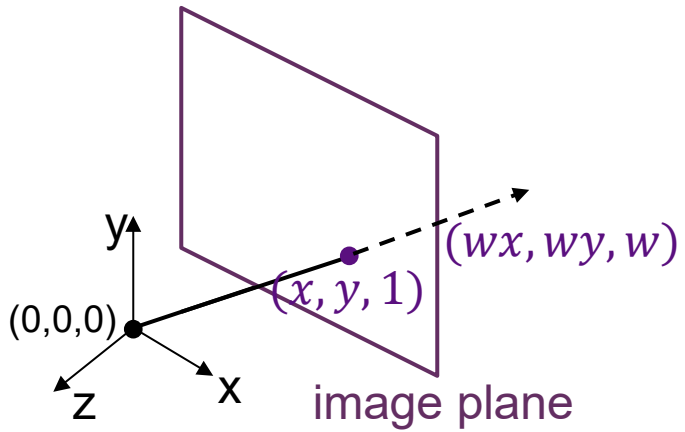
- represent coordinates in $\mathbb{P}^2$ with a 3-vector

$$\begin{bmatrix} x \\ y \end{bmatrix} \xrightarrow{\text{homogeneous coords}} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

← We will normally use w to denote this homogeneous vector element

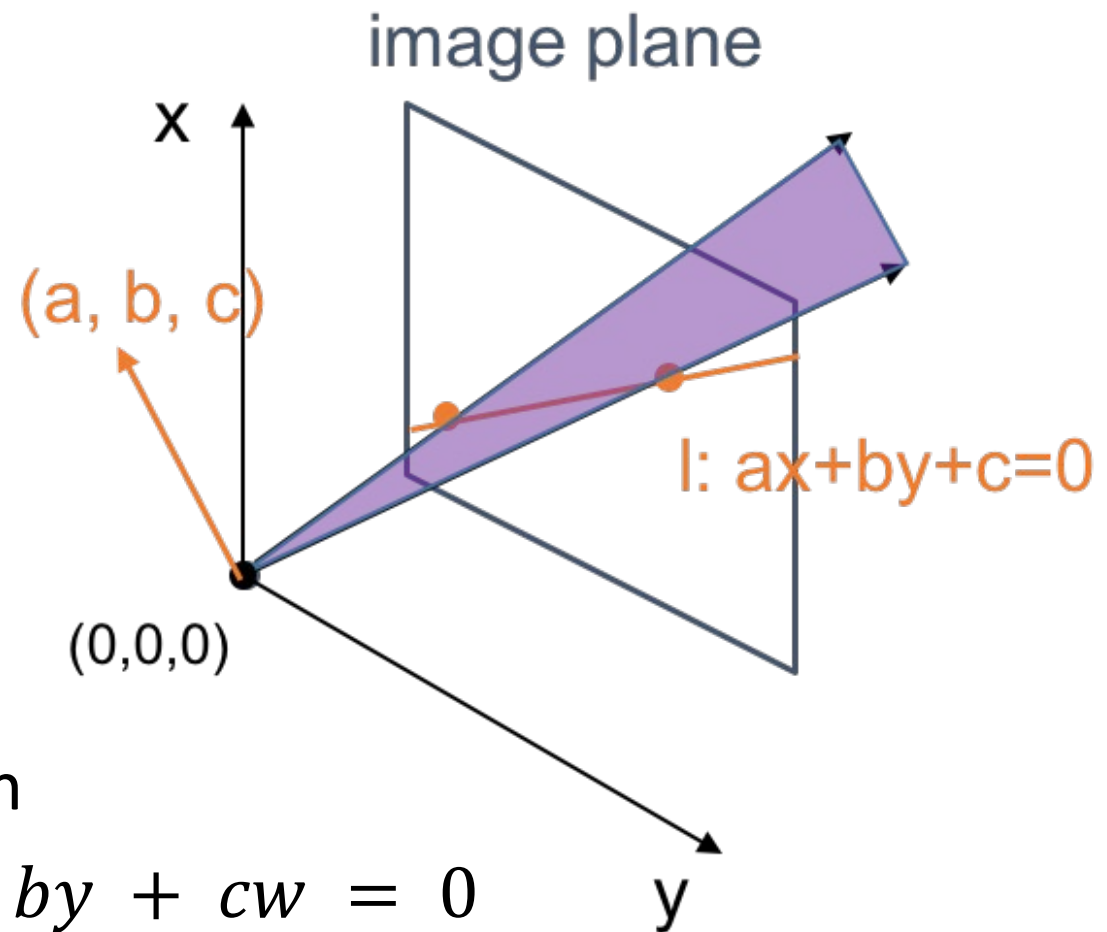
Recap: Projective $\mathbb{P}^n \rightarrow$ Euclidean $\mathbb{R}^n$

- $(x, y, w) = \left(\frac{x}{w}, \frac{y}{w}, 1\right)$, if $w \neq 0$
- Geometric intuition



- $(x, y, 0)$: an infinity point
Note: $\mathbb{P}^2$ does not contain $(0,0,0)$.

Recap: Projective lines



- A line is a *plane* of rays through origin
 - all rays (x, y, w) satisfying: $ax + by + cw = 0$

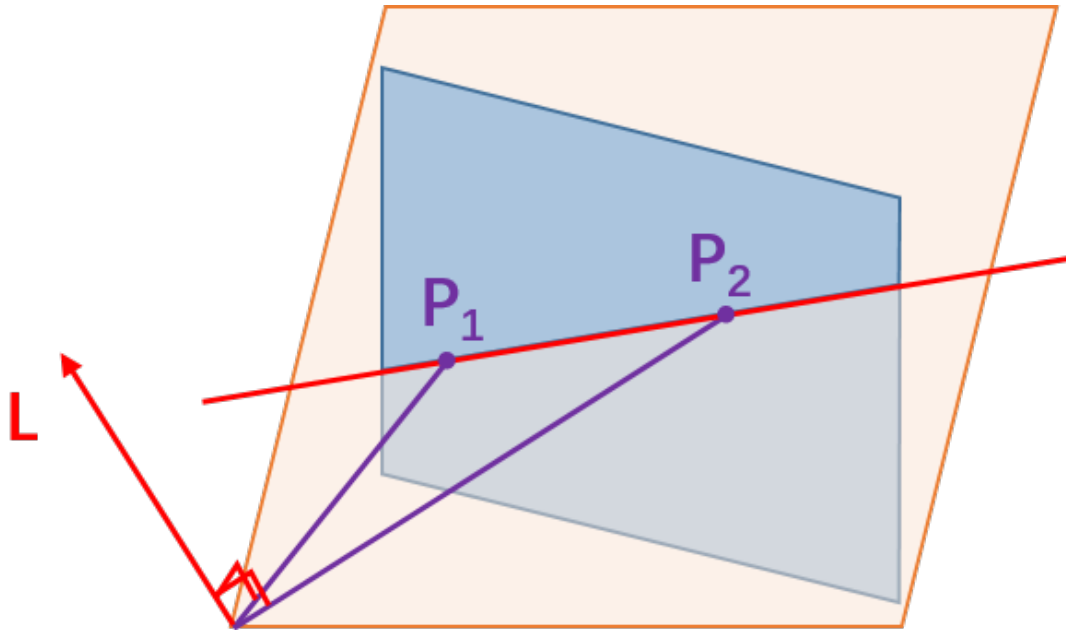
$$\begin{bmatrix} a & b & c \end{bmatrix} \begin{bmatrix} x \\ y \\ w \end{bmatrix} = 0$$

$$l^T x = 0$$

Recap: Relationship btw Line and Point Representations in $\mathbb{P}^2$

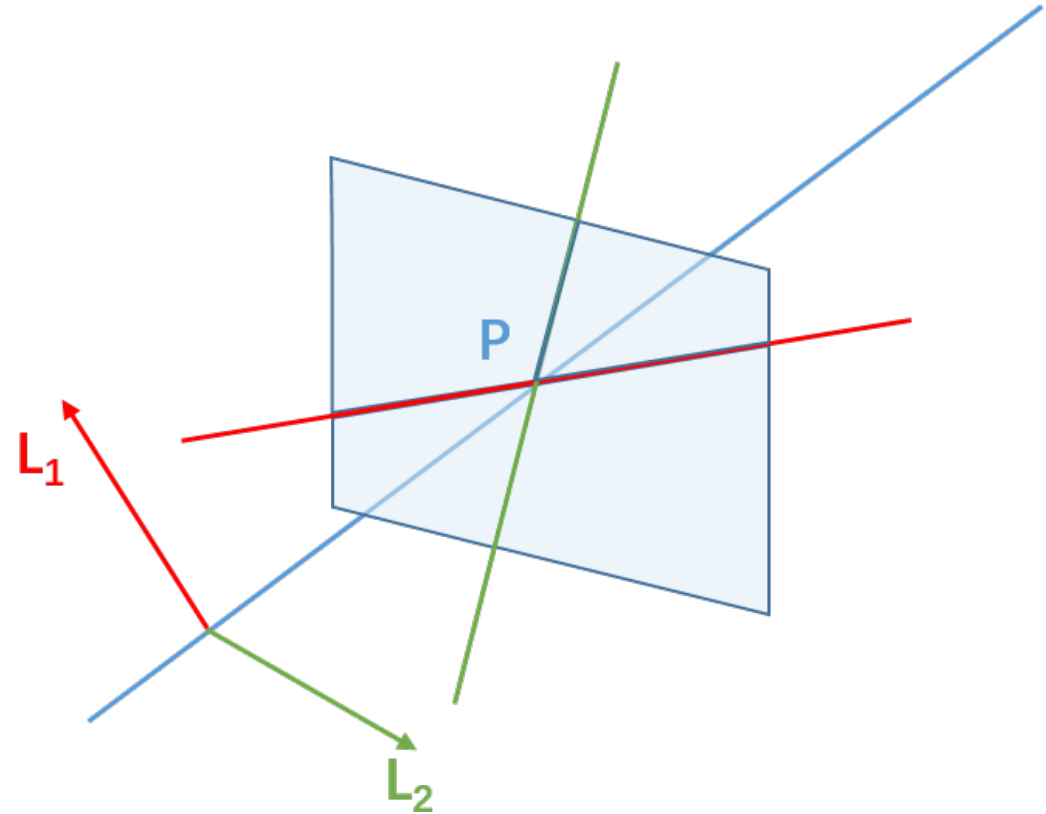
Given two points: $\mathbf{p}_1$ and $\mathbf{p}_2$

Define a line $\mathbf{l} = \mathbf{p}_1 \times \mathbf{p}_2$



Given two lines: $\mathbf{l}_1$ and $\mathbf{l}_2$

Define a point $\mathbf{p} = \mathbf{l}_1 \times \mathbf{l}_2$



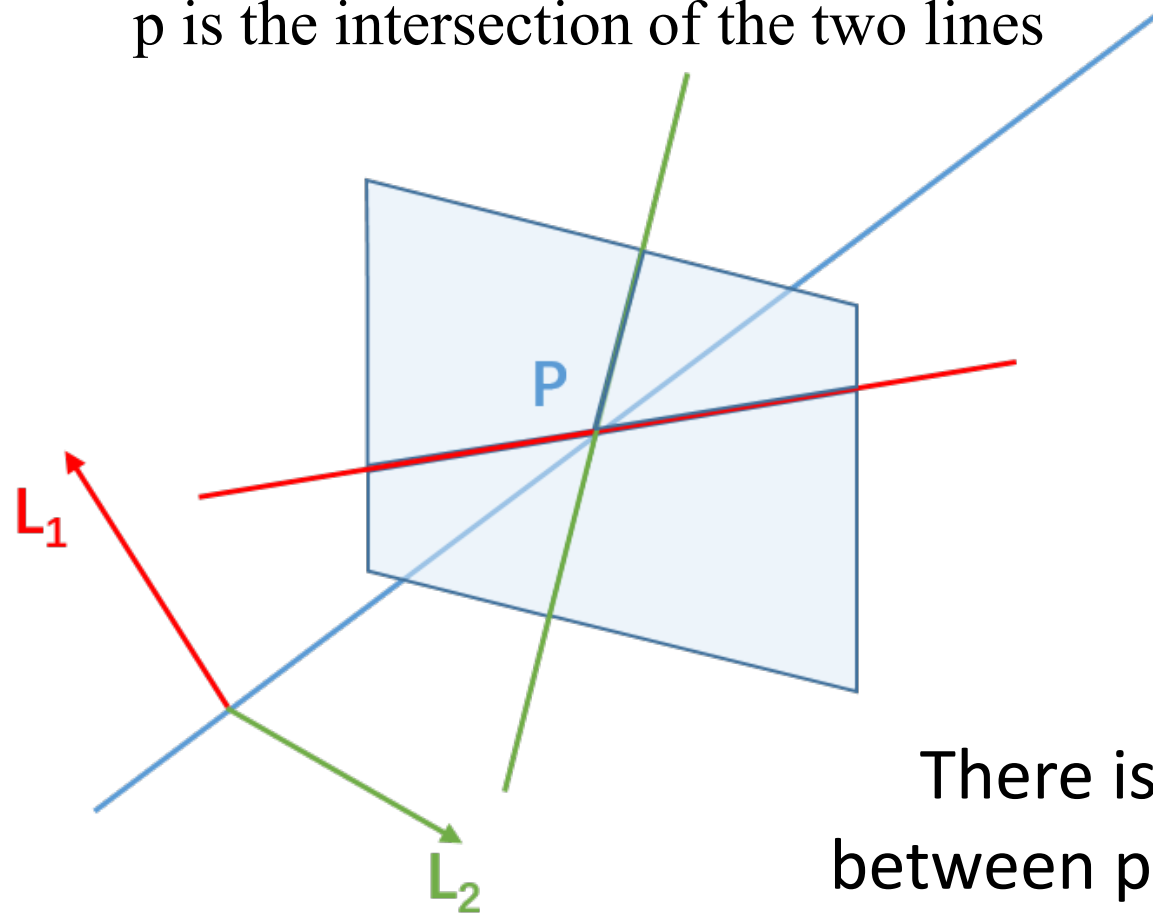
Intersection point of two lines in $\mathbb{P}^2$

Given two lines: l_1 and l_2

Define a point $p = l_1 \times l_2$

p is the intersection of the two lines

Q: Why? Hint: solve for the intersection by finding points that satisfy both line equations.



There is a close symmetry between points and lines in $\mathbb{P}^2$!

Exercise

- Where do the lines $x = 1$ and $x = 2$ on image plane intersect?

Extending to Any Parallel Lines

$$l = (a, b, c) \qquad l' = (a, b, c')$$

Intersection:

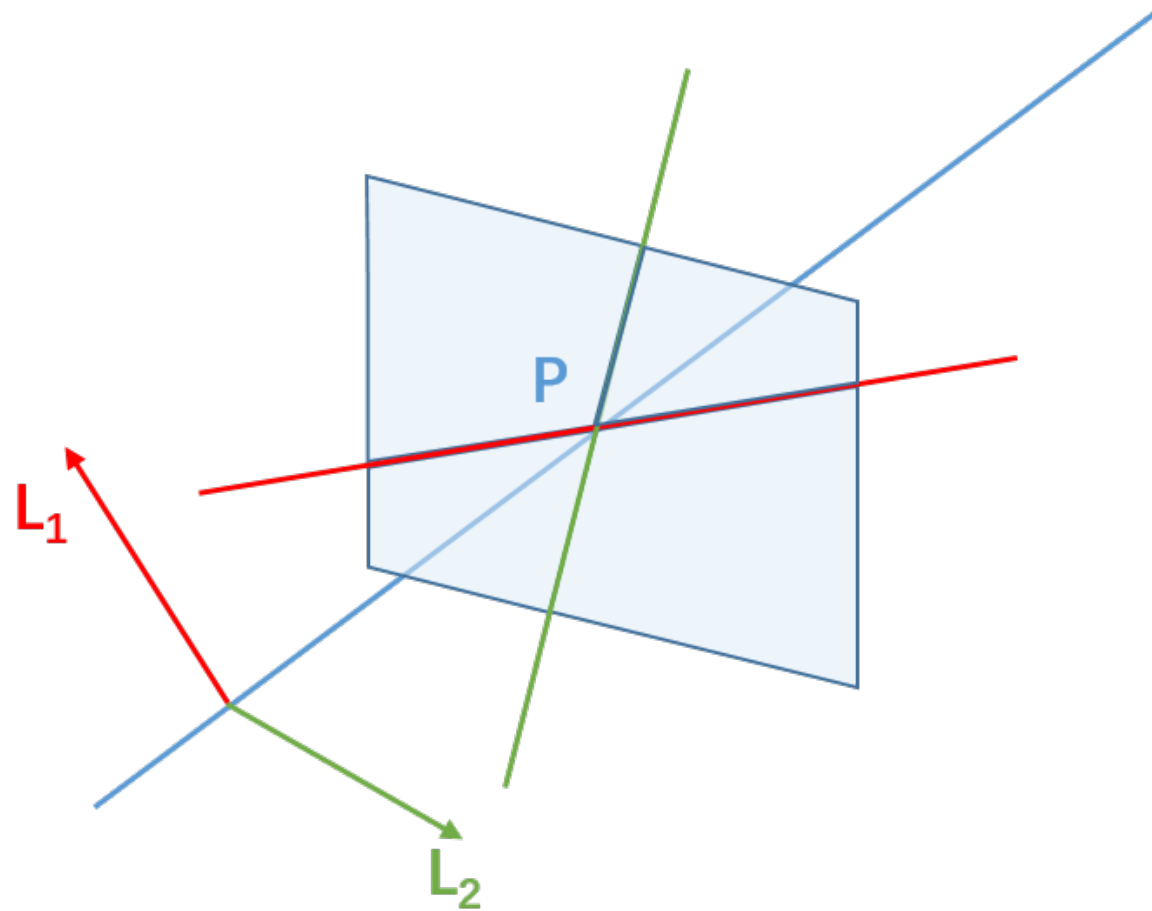
$$\begin{aligned} l \times l' &= l \times l' \\ &= \begin{vmatrix} i & j & k \\ a & b & c \\ a & b & c' \end{vmatrix} \\ &= (bc' - bc, ca - c'a, ab - ab)^T \\ &= (c' - c)(b, -a, 0)^T \end{aligned}$$

Any point $(x_1, x_2, 0)$ is intersection of parallel lines

Extending to Any Parallel Lines

- Under projective geometry,
 - All parallel lines intersect at a point at infinity
 - One point at infinity $\Leftrightarrow$ one parallel line direction
line $l = (a, b, c)^T$ intersects at $(b, -a, 0)^T$

Intersection point of two lines in $\mathbb{P}^2$

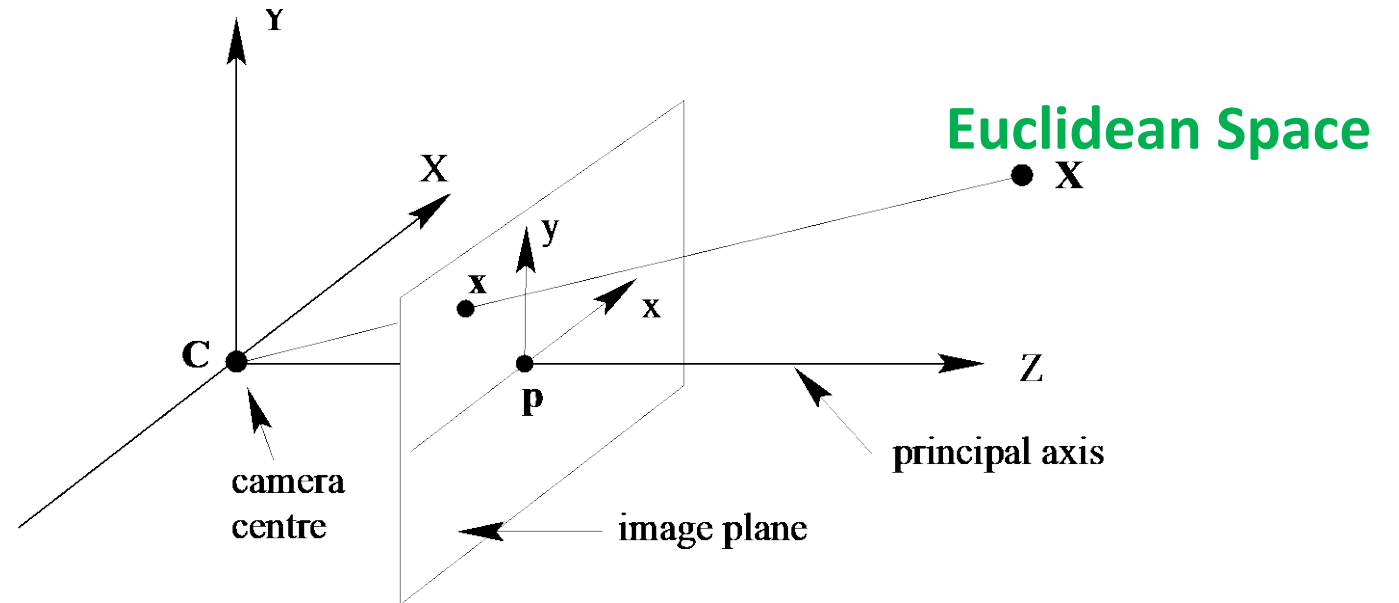
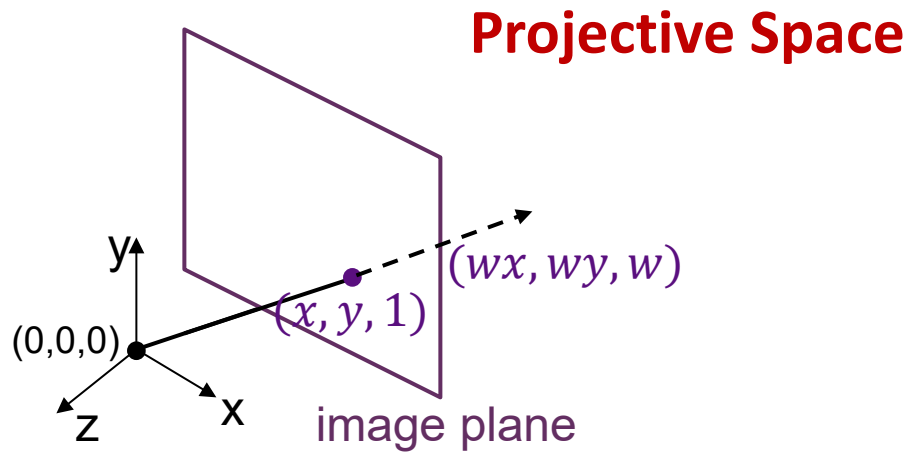


When does P have the form $(x,y,0)$?

Projective geometry $\leftrightarrow$ Euclidean interpretation

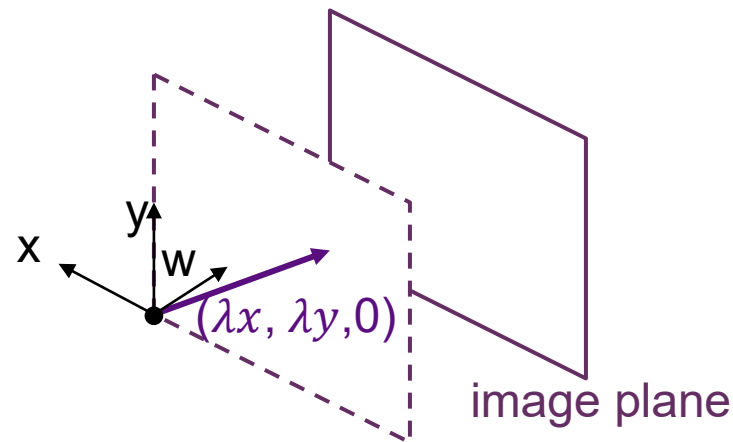
In the Euclidean interpretation, we treat w as the third spatial coordinate.

- The w axis is a scaled version of the principal axis Z (in camera-centric coordinates).
- The image plane is $w = 1$, same as $Z = f$
- $w = 0$ is the same as $Z = 0$. Parallel to image plane, passing through camera center.



Point at infinity / “ideal” points

- Ideal point (“point at infinity”)
 - $p \cong (x, y, 0)$ – rays through camera center parallel to image plane
 - It has infinite image coordinates

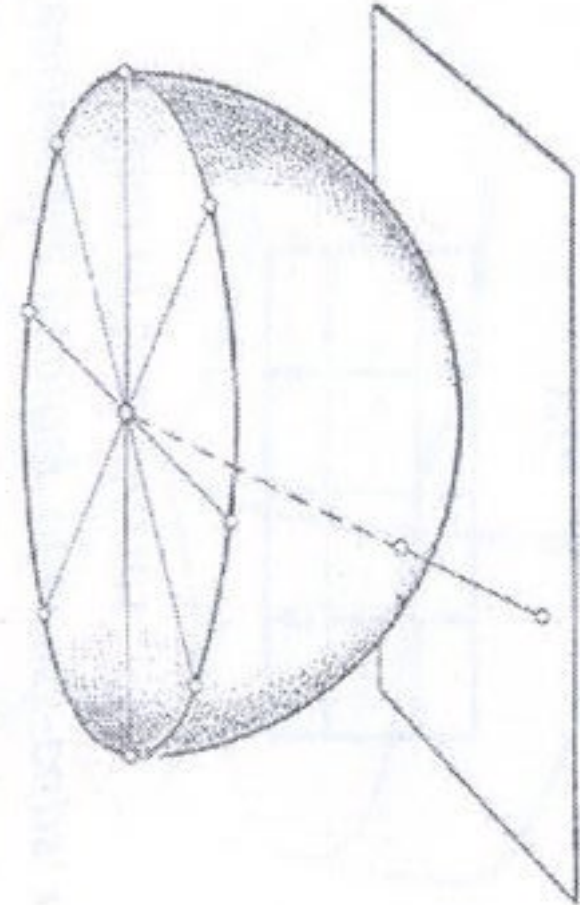


Point at infinity / “ideal” points

$(x_1, x_2, 0)$

Looking-at direction

“Ideal” points



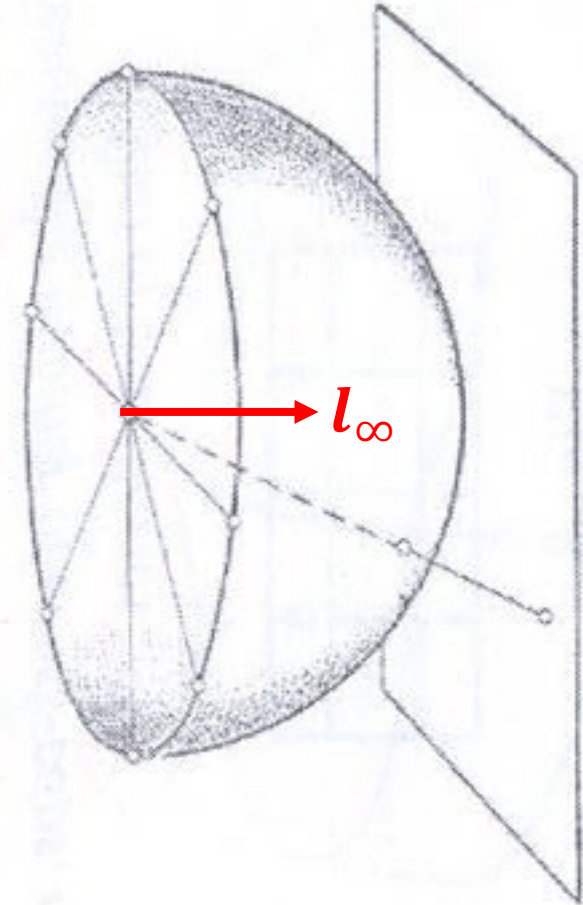
“Line at infinity”

- A line passing through all ideal points i.e. point

$$l_{\infty} = (0, 0, 1)^T$$

- Because :

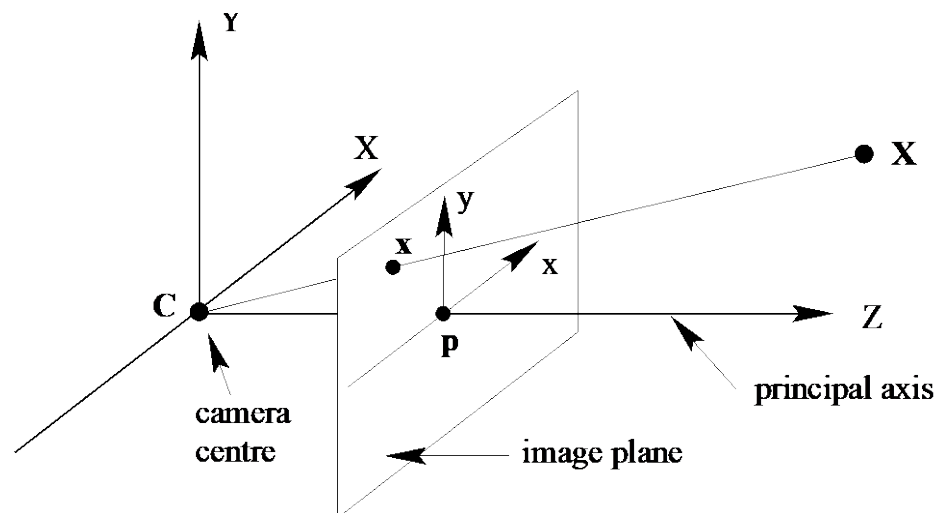
$$\begin{bmatrix} 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ 0 \end{bmatrix} = 0$$



Perspective Projections are Linear in $\mathbb{P}$

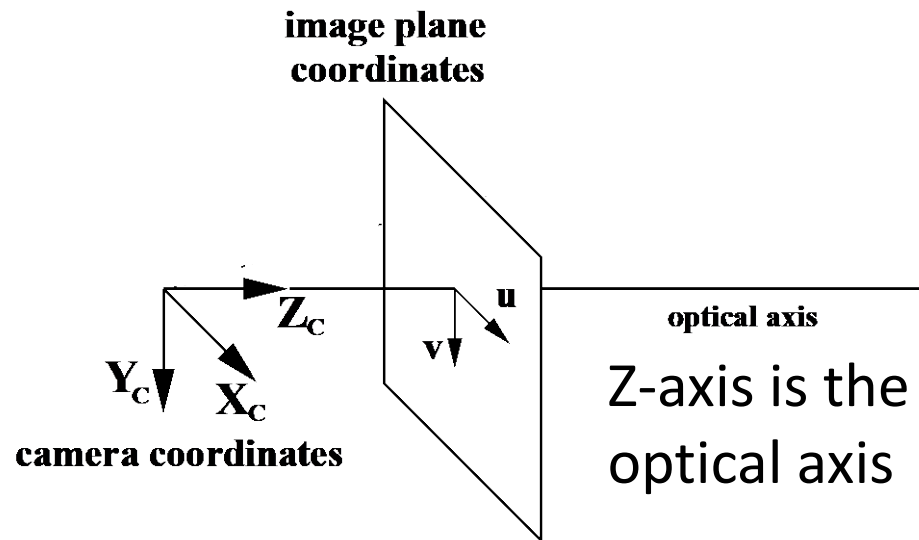
Recall:

$$x = f \frac{X}{Z}, y = f \frac{Y}{Z}$$



$$\begin{bmatrix} X \\ Y \\ Z \\ 1 \end{bmatrix} \rightarrow \begin{bmatrix} fX \\ fY \\ Z \end{bmatrix}$$

Camera Coordinate System + Principal Point Offset



The image plane (u, v) is perpendicular to the optical axis.

Intersection of the image plane with the optical axis is the *image center* (u_o, v_o)

Projection
in pixels

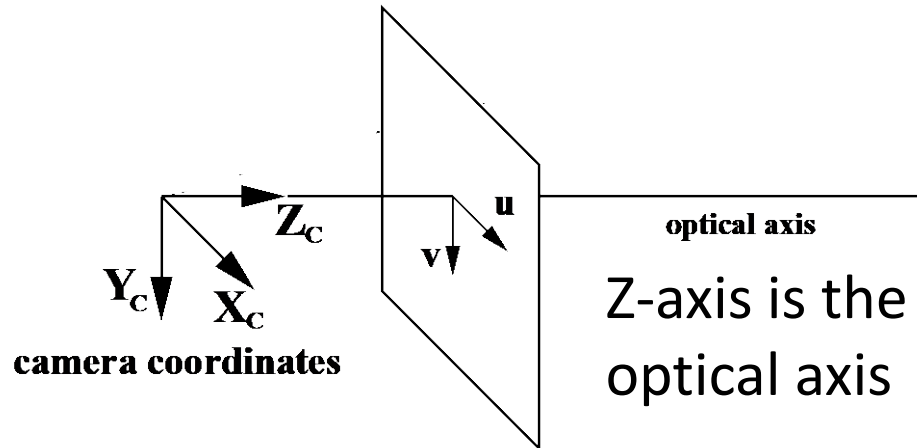
$$u = f \frac{X_c}{Z_c} + u_0,$$

$$v = f \frac{Y_c}{Z_c} + v_0$$

Projection equation with image origin $\neq$ principal point

$$u = f \frac{X_c}{Z_c} + u_0, \quad v = f \frac{Y_c}{Z_c} + v_0$$

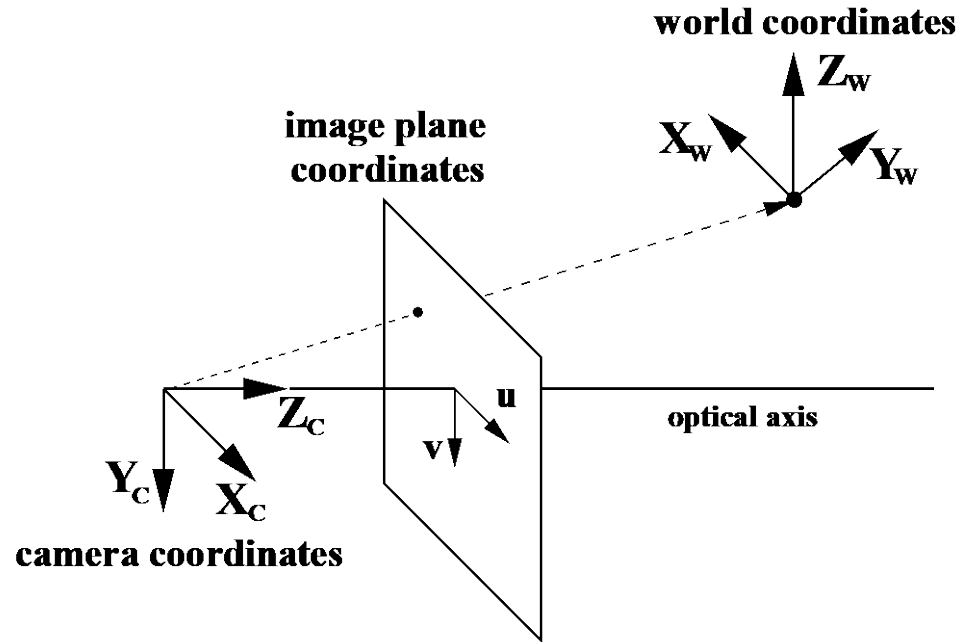
image plane
coordinates



Bold face to indicate this is the full vector in $\mathbb{P}^3$

$$\lambda \begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = \begin{bmatrix} f & & u_0 \\ & f & v_0 \\ & & 1 \end{bmatrix} \begin{bmatrix} 1 & & 0 \\ & 1 & 0 \\ & & 1 \end{bmatrix} \begin{bmatrix} X_c \\ Y_c \\ Z_c \\ 1 \end{bmatrix} = K[I|0] \mathbf{X}_c$$

From world to camera: Euclidean transformation



$$\begin{bmatrix} X_c \\ Y_c \\ Z_c \end{bmatrix} = R_{3 \times 3} \begin{bmatrix} X_w \\ Y_w \\ Z_w \end{bmatrix} + t = \begin{bmatrix} R_{3 \times 3} & t \\ 0 & 1 \end{bmatrix} X_w$$

What does R mean exactly?

Think “change of coordinates”.

- The camera has its own axes (X_c, Y_c, Z_c)
- The world has its axes (X_w, Y_w, Z_w)

R tells you how to express world vectors in the camera basis.

That is, R denotes the rotation of the world axes w.r.t. camera axes

= *inverse* rotation of the camera axes w.r.t. world axes.

What about t ?

Let C be the camera center in world coordinates.

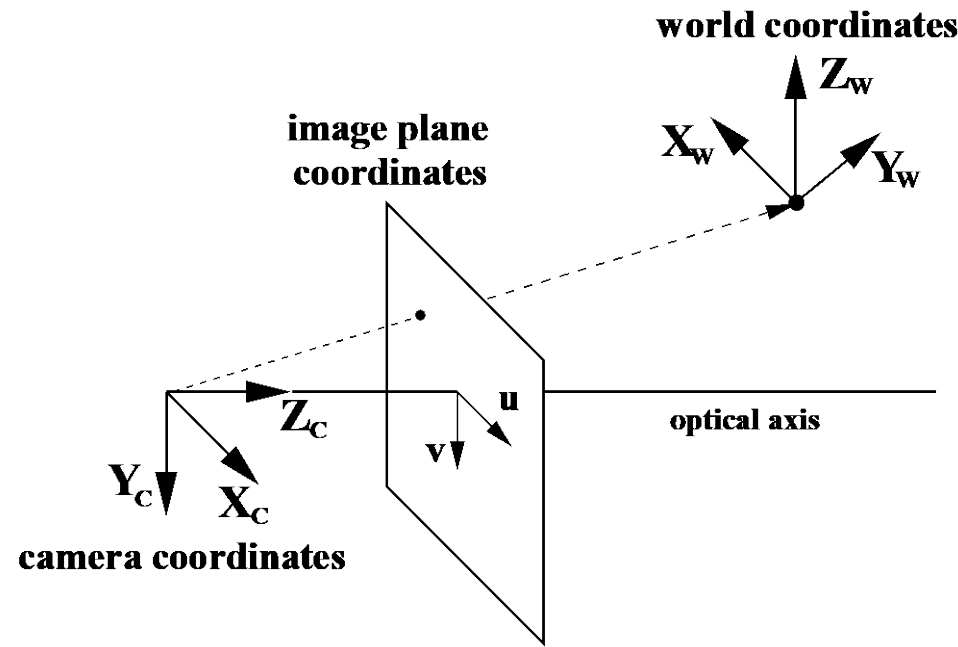
That means: C is the world point that maps to the camera origin $(0, 0, 0)$.

Plug $X_w = C$ into the equation:

$$0 = R C + t \Rightarrow t = -R C.$$

t here is actually R times the translation of world origin from camera origin.

Putting the pieces together: Projection matrix P



camera 3D coords to pixels Convert world to camera coordinates

$$\lambda \begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = \begin{bmatrix} f & & u_0 \\ & f & v_0 \\ & & 1 \end{bmatrix} \begin{bmatrix} R_{3 \times 3} & \mathbf{t} \end{bmatrix} \begin{bmatrix} X_w \\ Y_w \\ Z_w \\ 1 \end{bmatrix} = K[R|\mathbf{t}]\mathbf{X}_w$$

Summarizing

- Permitting principal point offset (i.e. shifted origin for image plane)

$$\begin{bmatrix} X \\ Y \\ Z \\ 1 \end{bmatrix} \rightarrow \begin{bmatrix} fX + Zu_0 \\ fY + Zv_0 \\ Z \\ 1 \end{bmatrix} = \underbrace{\begin{bmatrix} f & u_0 & 0 \\ & f & v_0 & 0 \\ & & 1 & 0 \end{bmatrix}}_{\text{"Intrinsics" } K} \begin{bmatrix} X \\ Y \\ Z \\ 1 \end{bmatrix} = K[I|\mathbf{0}]\mathbf{X}_c$$

- The above assumes X, Y, Z in camera-centric coordinates. Permitting different “world” and “camera” coordinate frames:

$$\mathbf{X}_c = \underbrace{R_{3 \times 3}}_{\text{Rotate}} \begin{bmatrix} X \\ Y \\ Z \\ 1 \end{bmatrix} + \underbrace{\mathbf{t}}_{\text{Translate}} = [R|\mathbf{t}]\mathbf{X}_w,$$

$$\mathbf{x} \sim K[R|\mathbf{t}]\mathbf{X}_w$$

First convert to camera coordinates

Then project as before.

Perspective Projection in homogeneous coordinates

$$\begin{bmatrix} x \\ y \\ w \end{bmatrix} = P_{3 \times 4} \begin{bmatrix} X \\ Y \\ Z \\ 1 \end{bmatrix}$$

$$\underbrace{P_{3 \times 4}}_{\substack{\text{"Camera} \\ \text{projection} \\ \text{matrix"}}} = \underbrace{K_{3 \times 3}}_{\text{"Intrinsics"}} \underbrace{[R_{3 \times 3} | \mathbf{t}_{3 \times 1}]}_{\text{"Extrinsics"}}$$

Q: How many degrees of freedom does P have given the type of intrinsics we have assumed?

